

Regularization path, Pareto frontier and sparsity

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Laboratoire d'Informatique, de Traitement de l'Information et des Systèmes

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Plan

- 1 The L_2 regularization path and its Pareto frontier
- 2 Efficient computation requires sparsity
- 3 The LARS as a regularization path algorithm
- 4 Conclusion

Example: the (linear) Ridge Regression

► problem: $\min_{\beta \in \mathbb{R}^d} \|\mathbf{y} - \mathbf{X}\beta\|^2 + \lambda \|\beta\|^2$

► solution: $\beta(\lambda) = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$

$\mathbf{I}$ is the identity matrix

Regularization path $\mathcal{S}$: set of solutions as a function of λ

$$\mathcal{S} = \{\beta(\lambda) \mid \lambda \in [0, \infty[\}$$

also called the *solution path*

1D Ridge Regression in the costs domain

the Loss term L as a function of the penalty term P

$$\min_{\beta \in \mathbb{R}} \sum_{i=1}^n (x_i \beta - y_i)^2 + \lambda \beta^2$$

$$\begin{cases} L(\beta) = \sum_{i=1}^n (x_i \beta - y_i)^2 \\ P(\beta) = \beta^2 \end{cases}$$

$$\begin{cases} L(P) = aP \pm b\sqrt{P} + c \\ a, b \text{ and } c \in \mathbb{R} \end{cases}$$

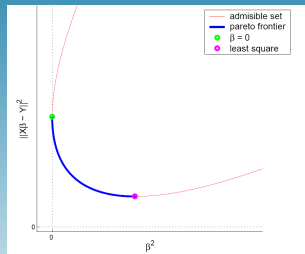


Figure: regularization path as a function of the criteria L and P .

Ridge Regression and dominance

$$\begin{cases} L(\beta) = \|y - X\beta\|^2 \\ P(\beta) = \|\beta\|^2 \end{cases}$$

dominance

a vector β_1 dominates another vector β_2 if $L(\beta_1) \leq L(\beta_2)$ and $P(\beta_1) \leq P(\beta_2)$

Pareto frontier

it is the set of all non dominated solutions

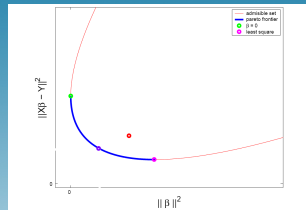


Figure: dominated point (red), non dominated point (purple) and Pareto frontier (blue).

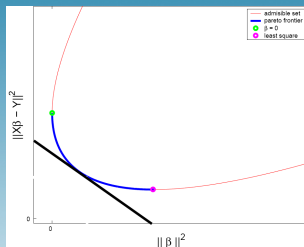
Pareto frontier $\Leftrightarrow$ Reg. path

3 equivalent formulations

$$\min_{\beta \in \mathbb{R}^d} \|y - X\beta\|^2 + \lambda \|\beta\|^2$$

$$\begin{cases} \min_{\beta} \|y - X\beta\|^2 \\ \text{s.t. } \|\beta\|^2 \leq C \end{cases}$$

$$\begin{cases} \min_{\beta} \|\beta\|^2 \\ \text{s.t. } \|y - X\beta\|^2 \leq C' \end{cases}$$



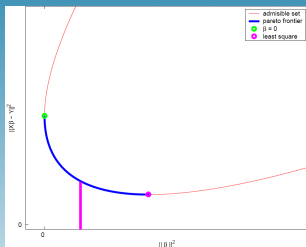
it works for CONVEX criteria!

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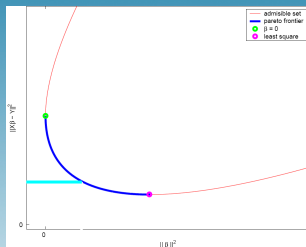
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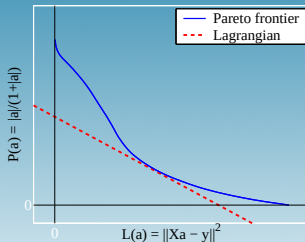
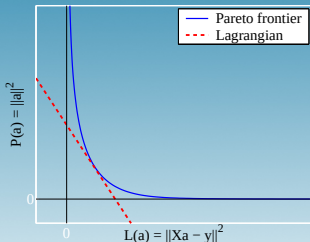
$$\begin{cases} \min_{\beta} \|\beta\|^2 \\ \text{s.t. } \|y - X\beta\|^2 \leq C' \end{cases}$$



it works for CONVEX criteria!

the importance of being convex

$$\begin{cases} \min_{\beta} L(\beta) \\ \min_{\beta} P(\beta) \end{cases} \quad \min_{\beta \in \mathbb{R}^d} L(\beta) + \lambda P(\beta)$$



in the non convex case

the 3 formulations are not equivalent

so far...

- ▶ learning is a multi objective problem
- ▶ the regularization path is the Pareto frontier
- ▶ beware the non convex case
- ▶ it works for more than 2 criteria

what for?

to tune (*efficiently*) the regularization parameter λ

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How to tune the regularization parameter λ ?

► brute force

for each $\lambda_1 < \lambda_2 < \dots < \lambda_k < \dots < \lambda_K$

compute $\beta_k = (X^\top X + \lambda_k I)^{-1} X^\top \mathbf{y}$, $k = 1, K$

$\mathcal{O}(Kn^3)$

► warm start

$\beta_k = \Phi(\beta_{k-1})$ (using ℓ conjugate gradient iterations)

$\mathcal{O}(K\ell n^2)$

► warm start + prediction step

$\beta_k^{(p)} = \beta_{k-1} + \rho \nabla_{\beta} (L(\beta_{k-1}) + \lambda_k P(\beta_{k-1}))$ (prediction step)

$\beta_k = \Phi(\beta_k^{(p)})$ (correction step using conjugate gradient)

$\mathcal{O}(K\ell' n^2)$

► use only the prediction step!

$\beta_k = \beta_{k-1} + \lambda_k \Psi(\beta_{k-1})$ (prediction step)

to do so the regularization path has to be piecewise linear

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to do so the regularization path has to be piecewise linear

$\mathcal{O}(Kn^2)$

How to choose L and P to get linear reg. path?

Solution path is linear $\Leftrightarrow$ one cost is piecewise quadratic and the other one piecewise linear

convex case [Rosset & Zhu, 07]

$$\min_{\beta \in \mathbb{R}^d} L(\beta) + \lambda P(\beta)$$

1. Piecewise linearity: $\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = \text{constant}$

2. optimality

$$\nabla L(\beta(\lambda)) + \lambda \nabla P(\beta(\lambda)) = 0$$

$$\nabla L(\beta(\lambda + \varepsilon)) + (\lambda + \varepsilon) \nabla P(\beta(\lambda + \varepsilon)) = 0$$

3. use Taylor expansion

$$\lim_{\varepsilon \rightarrow 0} \frac{\beta(\lambda + \varepsilon) - \beta(\lambda)}{\varepsilon} = [\nabla^2 L(\beta(\lambda)) + \lambda \nabla^2 P(\beta(\lambda))]^{-1} \nabla P(\beta(\lambda))$$

$$\nabla^2 L(\beta(\lambda)) = \text{constant} \quad \text{and} \quad \nabla^2 P(\beta(\lambda)) = 0$$

Piecewise linear regularization path algorithms

L	P	<i>regression</i>	<i>classification</i>	<i>clustering</i>
L_2	L_1	Lasso/LARS	L1 L2 SVM	OC SVM
L_1	L_2	SVR	SVM	
L_1	L_1	L1 least absolute deviation	L1 SVM	

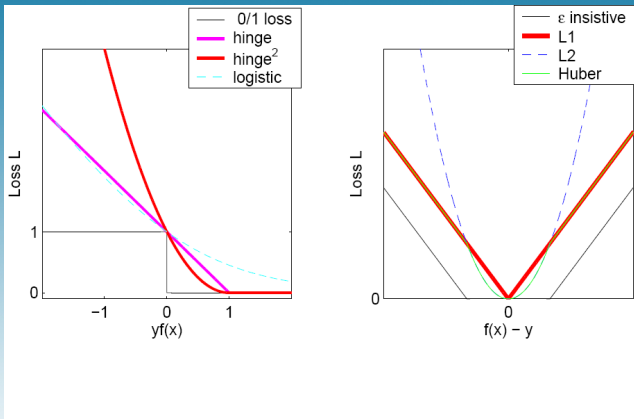
Table: example of piecewise linear regularization path algorithms.

$$P: L_p = \sum_{j=1}^d |\beta_j|^p \quad L: L_p: |f(\mathbf{x}) - y|^p \quad \text{hinge } (yf(\mathbf{x}) - 1)_+^p$$

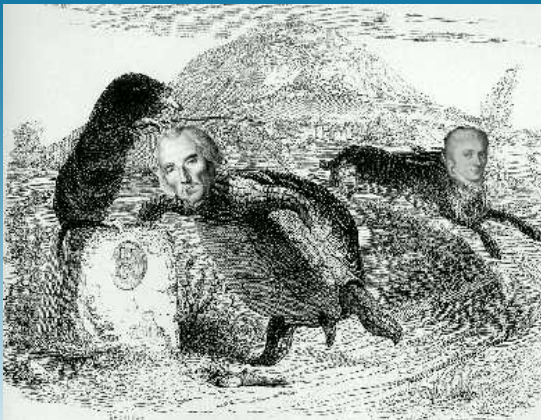
$$\varepsilon\text{-insensitive} \quad \begin{cases} 0 & \text{if } |f(\mathbf{x}) - y| < \varepsilon \\ |f(\mathbf{x}) - y| - \varepsilon & \text{otherwise} \end{cases}$$

$$\text{Huber's loss:} \quad \begin{cases} |f(\mathbf{x}) - y|^2 & \text{if } |f(\mathbf{x}) - y| < t \\ 2t|f(\mathbf{x}) - y| - t^2 & \text{otherwise} \end{cases}$$

Examples of Loss



the world is changing



The Gaussian Hare and the Laplacian Tortoise
Computability of L_1 vs. L_2 Regression Estimators.

Portnoy & Koenker 1997

The consequence of having useful reg. path

$$\min_{\beta \in \mathbb{R}^d} L(\beta) + \lambda P(\beta) \quad \Leftrightarrow \quad \{\beta(\lambda) \mid \lambda \in [0, \infty]\}$$

- ▶ efficient computing

$\Rightarrow$ piecewise linearity

$$\beta^{\text{NEW}} = \beta^{\text{OLD}} + (\lambda^{\text{NEW}} - \lambda^{\text{OLD}})u$$

- ▶ piecewise linearity

$\Rightarrow$ either L or P is L_1

- ▶ L_1 criteria

$\Rightarrow$ sparsity: *a lot of* $\beta_j = 0$

sparsity and active constraints

why does L_1 provide sparsity?

Definition: strong homogeneity set (variables)

$$l_0 = \{j \in \{1, \dots, d\} \mid \beta_j = 0\}$$

Theorem

Regular if $L(\beta) + \lambda P(\beta)$ differentiable and if $l_0(\mathbf{y}) \neq \emptyset$

$$\forall \varepsilon > 0, \exists \mathbf{y}' \in \mathcal{B}(\mathbf{y}, \varepsilon) \text{ such that } l_0(\mathbf{y}') \neq l_0(\mathbf{y})$$

Singular if $L(\beta) + \lambda P(\beta)$ **NON**differentiable and if $l_0(\mathbf{y}) \neq \emptyset$

$$\exists \varepsilon > 0, \forall \mathbf{y}' \in \mathcal{B}(\mathbf{y}, \varepsilon) \text{ then } l_0(\mathbf{y}') = l_0(\mathbf{y})$$

singular criteria $\implies$ sparsity

Nikolova, 2000

L_1 criteria are singular in 0

singularity provides sparsity

Lasso and Basis pursuit

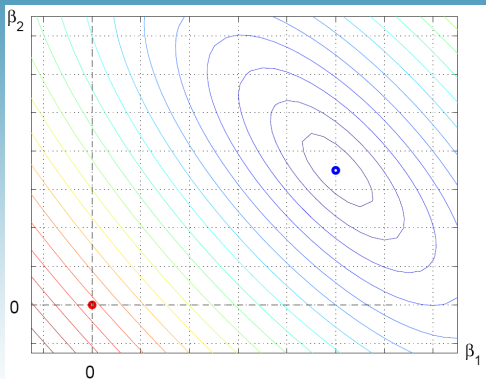
$$\min_{\beta \in \mathbb{R}^d} \|\mathbf{y} - X\beta\|^2 + \lambda \sum_{i=1}^d |\beta_i| \iff \begin{cases} \min_{\beta \in \mathbb{R}^d} \|\mathbf{y} - X\beta\|^2 \\ \text{with } \sum_{i=1}^d |\beta_i| \leq t \end{cases}$$

$$\iff \begin{cases} \min_{\beta \in \mathbb{R}^d} \sum_{i=1}^d |\beta_i| \\ \text{with } \|\mathbf{y} - X\beta\|^2 \leq \epsilon \end{cases}$$

Tibshirani, 1996 ; Chen, Donoho & Saunders, 1999 (Basis pursuit);
Donoho *et al.*, 2004 - Tropp 2004

2d LASSO and sparsity

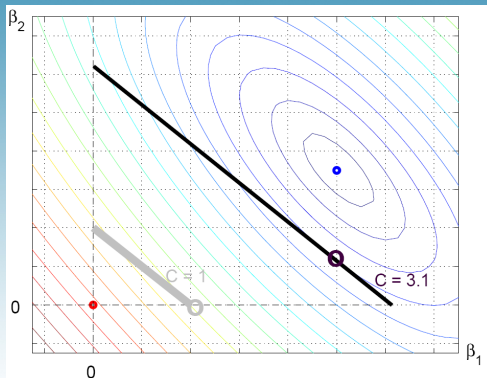
$$\begin{cases} \min_{\beta \in \mathbb{R}^2} L(\beta) = \|\mathbf{y} - X\beta\|^2 \end{cases}$$



Isocost lines
 $L = \text{Constant}$

2d LASSO and sparsity

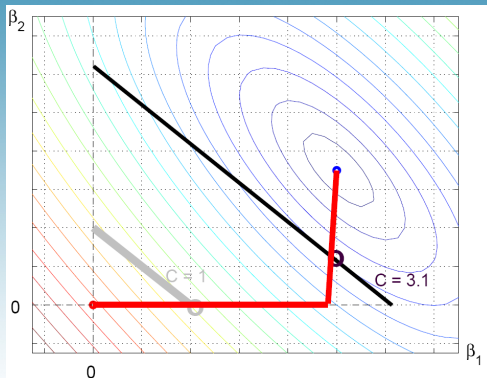
$$\begin{cases} \min_{\beta \in \mathbb{R}^2} & L(\beta) = \|\mathbf{y} - X\beta\|^2 \\ \text{with} & P(\beta) = |\beta_1| + |\beta_2| \leq C \end{cases}$$



$$C = 0 \ (\lambda = \infty)$$

2d LASSO and sparsity

$$\begin{cases} \min_{\beta \in \mathbb{R}^2} & L(\beta) = \|\mathbf{y} - X\beta\|^2 \\ \text{with} & P(\beta) = |\beta_1| + |\beta_2| \leq C \end{cases}$$



in red : the
piecewise linear
regularization path
 $\{\beta | C \in [0, C_{max}]\}$

regularization path algorithm

$$\min_{\beta \in \mathbb{R}^d} L(\beta) + \lambda P(\beta) \quad \Leftrightarrow \quad \{\beta(\lambda) \mid \lambda \in [0, \infty]\}$$

- efficient computation requires

$\Rightarrow$ a lot of $\beta_j = 0$

$$\beta^{\text{NEW}} = \beta^{\text{OLD}} + (\lambda^{\text{NEW}} - \lambda^{\text{OLD}})\mathbf{u}$$

- compute $\mathbf{u}$
- find out $I_0 = \{j \mid \beta_j = 0\}$

active set type optimization algorithm

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Notations

- all variables are centered - explicative variables are normalized

$$\sum_{i=1}^n y_i = 0 \quad \sum_{i=1}^n x_{ij} = 0 \quad \sum_{i=1}^n x_{ij}^2 = 1 \quad j = 1, d$$

- recall the least square

$$\text{loss: } L(\beta) = \|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - \hat{\mu}\|^2$$

$$\text{prediction: } \hat{\mu} = X\beta = \sum_{j=1}^d \mathbf{x}_j \hat{\beta}_j$$

$$X = \begin{bmatrix} \vdots \\ \mathbf{x}_1 \mathbf{x}_2 \dots \mathbf{x}_j \dots \mathbf{x}_d \\ \vdots \end{bmatrix}$$

solution:

$$\nabla L(\beta) = 0 \quad \Leftrightarrow \quad \underbrace{X^\top (\mathbf{y} - \hat{\mu})}_{\text{correlations}} = 0 \quad \Leftrightarrow \quad \beta_{\text{LS}} = (X^\top X)^{-1} X^\top \mathbf{y}$$

A forward selection strategy

$$\beta_0 = 0$$

$$\hat{\mu}_0 = X\beta_0 = 0$$

$$\beta_1 \leftarrow \beta_0 + \gamma_1 \mathbf{w}_1$$

$$\hat{\mu}_1 \leftarrow \hat{\mu}_0 + \gamma_1 \mathbf{x}_1$$

$$\beta_2 \leftarrow \beta_1 + \gamma_2 \mathbf{w}_2$$

$$\hat{\mu}_2 \leftarrow \hat{\mu}_1 + \gamma_2 \mathbf{u}_2$$

$$\mathbf{w}_1 =$$

$$0$$

$$\vdots$$

$$0$$

$$1 \leftarrow \hat{j}_1$$

$$0$$

$$\vdots$$

$$0$$

$$\vdots$$

$$0$$

$$a \leftarrow \hat{j}_1$$

$$0$$

$$\mathbf{w}_2 =$$

$$\vdots$$

$$0$$

$$b \leftarrow \hat{j}_2$$

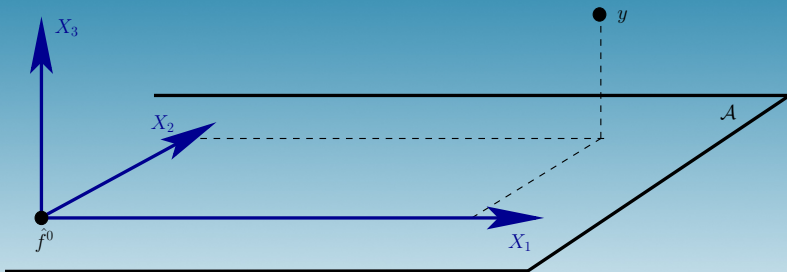
$$0$$

$$\vdots$$

l_β the set of interesting variables, l_0 set of $\beta_j = 0$

LAR: correlation and geometry (V. Guigue)

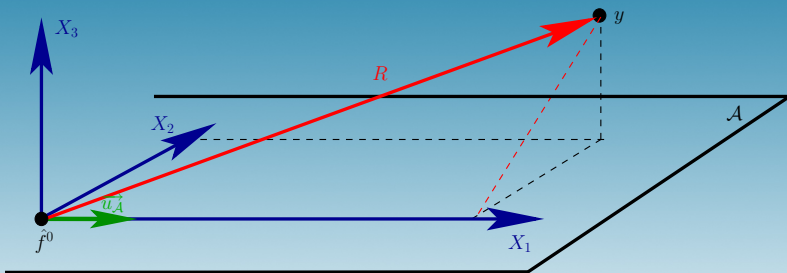
the solution in the X space



starting point: all the β are set to 0

LAR: correlation and geometry (V. Guigue)

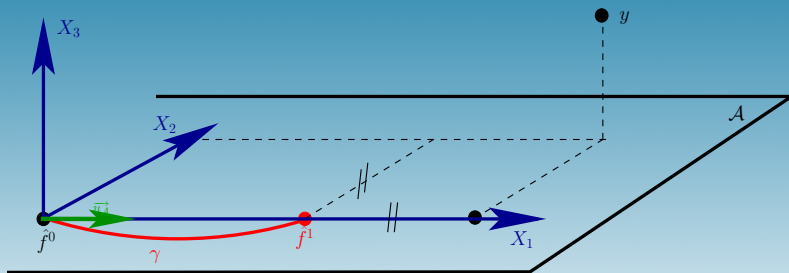
the solution in the X space



projection of the residual error on the active variable

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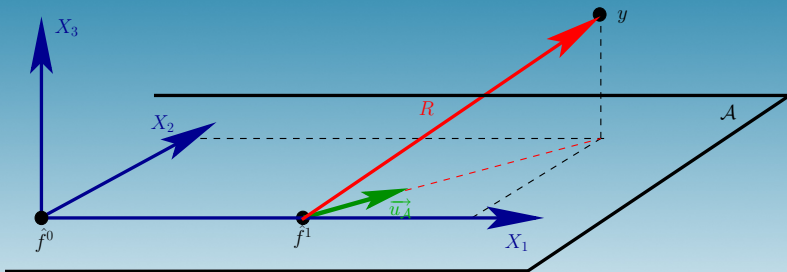
the solution in the X space



stepsize computation, same correlation of residual errors

LAR: correlation and geometry (V. Guigue)

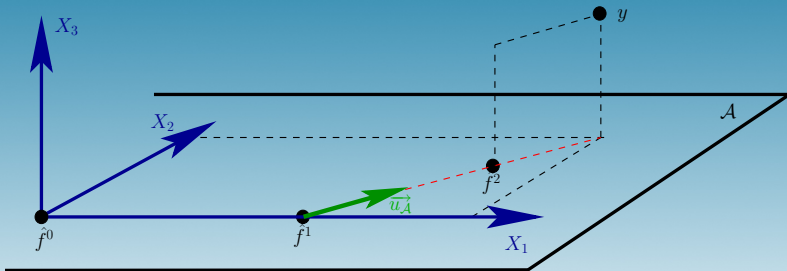
the solution in the X space



projection of the residual error on the active variable

LAR: correlation and geometry (V. Guigue)

the solution in the X space



stepsize computation

The LAR Algorithm

I_β = the set of *interesting* variables

$$\beta_{k+1} \leftarrow \beta_k + \gamma \mathbf{w}_{k+1} \in \mathbb{R}^d$$

$$\beta_{I_\beta} \leftarrow \beta_{I_\beta} + \gamma \mathbf{w}_{I_\beta} \in \mathbb{R}^{\text{card}(I_\beta) \leq d} \quad \beta_{I_\beta} = \beta_k(I_\beta)$$

$$\hat{\mu}_{k+1} \leftarrow \hat{\mu}_k + \gamma \mathbf{u}_{k+1} \in \mathbb{R}^n \quad \mathbf{u} = X\mathbf{w} = X_{I_\beta} \mathbf{w}_{I_\beta}$$

the LAR

1. find out the more interesting variable to add $j : I_\beta = I_\beta \cup \{j\}$
2. compute the descent direction ($\mathbf{w}, \mathbf{u} = X\mathbf{w} = X_{I_\beta} \mathbf{w}_{I_\beta}$)
3. compute the stepsize γ

The LAR Algorithm

$$\begin{aligned}\beta_{I_\beta} &\leftarrow \beta_{I_\beta} + \gamma_k \mathbf{w}_{I_\beta} & \{1, \dots, d\} &= I_\beta \cup I_0 \\ \hat{\mu}_{k+1} &\leftarrow \hat{\mu}_k + \gamma_k \mathbf{u}_k\end{aligned}$$

1. find **the variable to add** j : maximizing the gradient

$$j = \operatorname{argmax}_{j \in I_0} |\nabla_{\beta_{I_0}} S(\beta)| = |X_{I_0}^\top (\mathbf{y} - \hat{\mu})| \quad ; \quad I_\beta = I_\beta \cup j$$

2. compute the **descent direction**: second order approach

$$\mathbf{w}_{I_\beta} = H^{-1} \nabla_{\beta_{I_\beta}} L(\beta) \quad \left\{ \begin{array}{l} \nabla_{\beta_{I_\beta}} L(\beta) = X_{I_\beta}^\top (\mathbf{y} - \hat{\mu}) = c \mathbb{I}_{I_\beta} \\ H = X_{I_\beta}^\top X_{I_\beta} \end{array} \right.$$

$$(X_{I_\beta}^\top X_{I_\beta}) \mathbf{w}_{I_\beta} = \mathbb{I}_{I_\beta}$$

3. compute **the stepsize** γ

to maintain the same correlation: $X_{I_\beta}^\top (\mathbf{y} - \hat{\mu}) = c' \mathbb{I}$

what do we have to compute?

1. find the variable to add j : $\mathcal{O}(n(d - k))$

$$|X_{I_0}^\top (\mathbf{y} - \hat{\boldsymbol{\mu}})|$$

2. descent direction: Choleski update

$$X_{I_\beta} = X_{I_\beta} \cup s_j X_j \quad s_j = \text{sign}(\hat{c}_j)$$

$$(X_{I_\beta}^\top X_{I_\beta}) \mathbf{w}_{I_\beta} = \mathbb{I}_{I_\beta} \quad \mathcal{O}(nk + k^2)$$

3. compute the stepsize γ , $\mathcal{O}(n(d - k))$

$\Leftrightarrow$ this gives us the next variable to take into account

$$X_{I_\beta}^\top (\mathbf{y} - \hat{\boldsymbol{\mu}}) = c' \mathbb{I} \quad \Leftrightarrow \quad c' = \max_{j \in I_0} |\mathbf{x}_j^\top (\mathbf{y} - \hat{\boldsymbol{\mu}})|$$

the LAR at work (from [Efron *et al.*, 03])

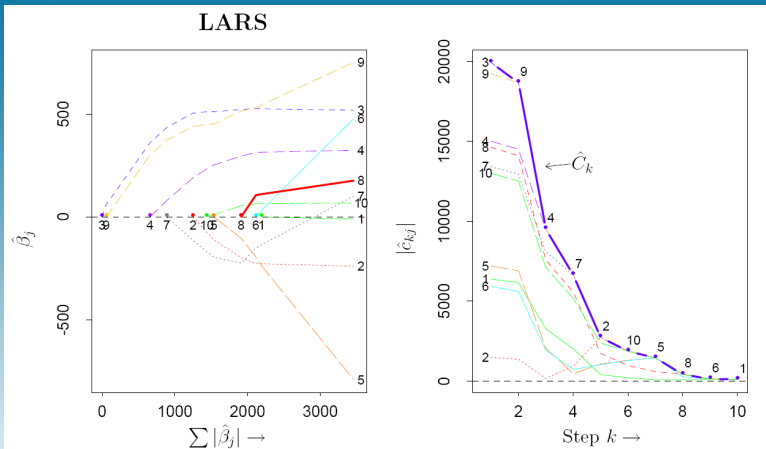


Figure 3. LARS analysis of the diabetes study. *Left:* estimates of regression coefficients $\hat{\beta}_j$, $j = 1, 2, \dots, 10$; plotted versus $\sum |\hat{\beta}_j|$; plot is slightly different than either Lasso or Stagewise, Figure 1. *Right:* Absolute current correlations as function of LARS step; variables enter active set (2.9) in order 3, 9, 4, 7, $\dots$, 1; heavy curve shows maximum current correlation $\hat{C}_k$ declining with k .

Adaptation of LAR to the LASSO

$$\left\{ \begin{array}{l} \min_{\beta \in \mathbb{R}^d} L(\beta) = \|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - \hat{\mu}\|^2 \\ \text{avec} \quad P(\beta) = \sum_{j=1}^d |\hat{\beta}_j| \leq t \end{array} \right.$$

if $\exists j$ such that β_j change sign

then γ is chosen such that $\beta_j = 0$ and this variable left l_β

Theorem 1

with the above adaptation

the LARS gives **the** LASSO solution path

regularization path - Theorem 7

$$\beta(t) = \beta(t_0) - (t - t_0) A_{I_\beta} \mathbf{w}_{I_\beta}$$

S is the sign

1. $\beta(t) = \operatorname{argmin} \|\mathbf{y} - X_{I_\beta} \beta\|^2$ with $\sum_{I_\beta} s_j \beta_j = t$
2. $\mathcal{L}(\beta, \lambda) = \|\mathbf{y} - X_{I_\beta} \beta\|^2 + \lambda (\sum_{I_\beta} s_j \beta_j - t)$
 $\nabla_{\beta} \mathcal{L}(\beta, \lambda) = X_{I_\beta}^\top (\mathbf{y} - X_{I_\beta} \beta) + \lambda S_{I_\beta} \mathbb{I} = 0$
3. $\nabla \mathcal{L}(t_1) - \nabla \mathcal{L}(t_2) = X_{I_\beta}^\top X_{I_\beta} (\beta(t_2) - \beta(t_1)) + (\lambda_1 - \lambda_2) S_{I_\beta} \mathbb{I}$
 $\Leftrightarrow \beta(t_2) - \beta(t_1) = (\lambda_1 - \lambda_2) S_{I_\beta} (X_{I_\beta}^\top X_{I_\beta})^{-1} \mathbb{I}$
4. $s_{I_\beta} (\beta(t_2) - \beta(t_1)) = t_2 - t_1 = (\lambda_1 - \lambda_2) A_{I_\beta}^{-1}$; $A_{I_\beta}^{-1} = \mathbb{I}^\top (X_{I_\beta}^\top X_{I_\beta})^{-1} \mathbb{I}$
5. $\beta(t_2) - \beta(t_1) = (t_1 - t_2) A_{I_\beta} S_{I_\beta} \underbrace{(X_{I_\beta}^\top X_{I_\beta})^{-1} \mathbb{I}}_{\mathbf{w}_{I_\beta}}$

it is piecewise linear

LAR(s) summary

- ▶ LARS is efficient
- ▶ LARS = LASSO, LARS = *stagewise*
- ▶ the whole regularization path
 - when to stop
- ▶ stability issue
 - you may prefer to use the *stagewise* LARS
 - the grouped lasso or the elastic net
- ▶ curse of dimensionality: LARS is ok up to $\text{card}(I_\beta) \leq 10^3$
 - work coordinate wise or randomise

2000 Homotopy algorithm for the LASSO [Osborne et al.]

2003 LARS: reg path algorithm for the LASSO [Efron et al.]

2004 SVM regularization path [Hastie et al.]

2004 piecewise linear theorem - curved regularization [Rosset & Zhu]

2004-06 prediction correction (glmpath) [Park et al., Friedman et al.]

2004-07 many particular cases

- ▶ boosting as a reg. path algorithm [Rosset et al.]
- ▶ elastic net [Zou & Hastie], Grouped LASSO [Yuan & Lin]
- ▶ doubly regularized SVM [Wang et al.], SVR [Gunter & Zhu]
- ▶ asymmetric costs for SVM [Bach et al.]
- ▶ Kernel LARS [Guigue et al.] ν SVM [Loosli et al.], ν SVR [Gasso et al.], OC SVM [Rakotomamonjy et al.], Laplacian SVM (semi supervised) [Gasso et al.]

2004-07 componentwise algorithms [Friedman et al.], Danzig selector [Candes & Tao]

the OC SVM regularization path [Davy et al.]

$$\left\{ \begin{array}{ll} \min_{\beta, \rho, \xi_i} & \frac{\lambda}{2} \|\beta\|^2 + \sum_{i=1}^n \xi_i - \lambda \rho \\ \text{such that} & \xi_i \geq 0 \quad \text{and} \quad \phi(X_i)^\top \beta \geq \rho - \xi_i \quad i = 1, n \end{array} \right.$$

- ▶ for a given λ : a density level set
- ▶ both α 's and ρ are piecewise linear in λ
- ▶ the regularization path gives the whole density

the OC SVM regularization path

provided by [A. Rakotomamonjy]

Conclusion

- ▶ Pareto frontier
- ▶ sparsity L_1
- ▶ efficient algorithm
- ▶ some limits
 - ▶ instability
 - ▶ large data sets
 - ▶ when to stop
 - ▶ non convexity

Perspectives

- ▶ more problems, more criteria, more applications

Conclusion

- ▶ Pareto frontier
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- ▶ efficient algorithm
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coupling with active set
 randomize
 derive relevant bounds
 piecewise linearisation

Perspectives

- ▶ more problems, more criteria, more applications

Questions?

questions?

LAR(s) (and svmopath) in R
Matlab

www-stat.stanford.edu/~hastie

asi.insa-rouen.fr/~vguigue/LARS.html

kernlab
SVR Matlab

cran.r-project.org/src/contrib/Descriptions/kernlab.html

asi.insa-rouen.fr/~arakotom

Danzig selector

www.l1-magic.org/

biblio

- ▶ Least Angle Regression. Least angle regression, Bradley Efron, ; Annals of statistics, vol. 32 (2), pp.407-499, 2004
- ▶ Piecewise Linear Regularized Solution Paths, Saharon Rosset, Ji Zhu. Annals of Statistics, to appear, 2007
- ▶ Local strong homogeneity of a regularized estimator, Mila Nikolova, SIAM Journal on Applied Mathematics, vol. 61, no. 2, pp. 633-658, 2000.
- ▶ Two-Dimensional Solution Path for Support Vector Regression, Gang Wang, Dit-Yan Yeung, Frederick H. Lochovsky, ICML, 2006
- ▶ Algorithmic linear dimension reduction in the l_1 norm for sparse vectors, A. C. Gilbert, M. J. Strauss, J. A. Tropp, and R. Vershynin, 2006 (submitted).
- ▶ A tutorial on support vector regression, A. Smola and B. Scholkopf, Statistics and Computing 14(3), pp 199-222, 2004
- ▶ The Dantzig selector: Statistical estimation when p is much smaller than n , Emmanuel Candes and Terence Tao Submitted to IEEE Transactions on Information Theory, June 2005.
- ▶ An iterative thresholding algorithm for linear inverse problems with a sparsity constraint, I. Daubechies, M. Defrise and C. De Mol, Comm. Pure Appl. Math, 57, pp.1413-1541, 2004.
- ▶ Pathwise coordinate optimization, Jerome Friedman, Trevor Hastie and Robert Tibshirani, technical report, May 2007.